Sample Numerical Practice Questions for Network Security

Part A: Historical Ciphers

Question 1 (Caesar Cipher)
Encrypt the plaintext `SECURITY` using a Caesar cipher with key $k = 7$. Use the mapping $A = 0, B = 1, \ldots, Z = 25$.

Question 2 (Caesar Decryption)
A ciphertext `KHOOR` is encrypted using a Caesar cipher with key $k = 3$. Decrypt the ciphertext and recover the plaintext.

Part B: One-Time Pad

Question 3
Alice wants to send the message `WILPNETWORK`. The one-time pad key is `XAWQMRCTKUL`. Use the rule

$$C = (M + K) \bmod 26$$

Compute the ciphertext.

Part C: Hill Cipher

Question 4
Encrypt the plaintext `HELP` using the Hill cipher with the key matrix

$$K = \begin{bmatrix} 6 & 1 \\ 9 & 5 \end{bmatrix}$$

Use $A = 0, \ldots, Z = 25$.

Question 5
Check whether the matrix

$$K = \begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix}$$

is invertible modulo 26.

Part D: GCD and Extended Euclidean Algorithm

Question 6
Compute $\gcd(145, 278)$ using the Euclidean algorithm.
Question 7
Use the Extended Euclidean Algorithm to find integers x, y such that

$$145x + 278y = \gcd(145, 278).$$

Part E: Multiplicative Inverse

Question 8
Find the multiplicative inverse of 31 modulo 43.
Question 9
Find the multiplicative inverse of 11 modulo 26.

Part G: Linear Congruential Generator

Question 10
Consider the LCG defined as

$$X_{n+1} = (9X_n + 5) \bmod 17$$

with seed $X_0 = 12$.
Compute the first 6 pseudo-random numbers.
Question 11
Determine the period of the above LCG.

Part H: RSA

Question 12
Let $p = 19$, $q = 23$.

1. Compute n and $\phi(n)$.
2. Choose $e = 13$. Verify $\gcd(e, \phi(n)) = 1$.
3. Compute the private key d.

Question 13
Using the RSA parameters from Question 16, encrypt the message $M = 9$.

Part I: Diffie-Hellman

Question 14
Let $p = 29$ and generator $g = 10$.

- Alice chooses private key $a = 16$.
- Bob chooses private key $b = 5$.

Compute:

1. Alice's public key
2. Bob's public key
3. Shared secret key

Part J: Theorem-Based Questions

Question 15
Use Euler's theorem to compute

$$7^{100} \mod 15$$

Question 16
Use Fermat's Little Theorem and use it to compute

$$3^{100} \mod 7$$
